



4.1 Sampling and data collection

Statistics and probability · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A sample is a subset of a population used to make inferences about the whole. Simple random sampling gives every member equal chance of selection; stratified sampling divides the population into groups (strata) and samples proportionally from each; systematic sampling selects every k th member from a list; convenience sampling uses readily available subjects, and quota sampling fills predetermined group sizes non-randomly. Poor sampling methods introduce bias, meaning the sample does not represent the population well.

Worked example: follow the reasoning

A school has 300 boys and 200 girls. A stratified sample of 50 students is needed. How many boys and girls should be sampled?

Total population=500, sample fraction= $50/500=1/10$.

Boys: $300 \times (1/10)=30$. Girls: $200 \times (1/10)=20$.

Quick check · multiple choice

Which sampling method involves selecting every 10th customer entering a shop?

A Simple random sampling

B Stratified sampling

C Systematic sampling

D Convenience sampling

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A company has 120 junior staff, 60 senior staff, and 20 managers (200 total). A stratified sample of 40 employees is to be selected. [6]
- (a) Calculate the number of employees that should be sampled from each group.
- (b) State one advantage of stratified sampling over simple random sampling in this context.
- (c) A researcher instead only surveys staff who pass by the front desk on a Monday morning. Name this sampling method and state one problem with it.

Final answer: _____



Answers and explanation

Question 1 [6]

- (a) Sample fraction = $40/200 = 1/5$. Juniors: $120/5 = 24$. Seniors: $60/5 = 12$. Managers: $20/5 = 4$.
(b) Stratified sampling ensures each subgroup (level of staff) is proportionally represented, reducing the risk that one group dominates the sample by chance.
(c) This is convenience sampling; it may be biased since staff who pass the front desk on a Monday morning are not representative of all staff (e.g. those working remotely or in other departments are excluded).

Quick check answer

C. Selecting at a fixed interval (every k th item) defines systematic sampling.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.